

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/329881153>

Isometric training and long-term adaptations: Effects of muscle length, intensity, and intent: A systematic review

Article in *Scandinavian Journal of Medicine and Science in Sports* · January 2019

DOI: 10.1111/sms.13375

CITATIONS

134

READS

38,737

4 authors:



Dustin J Oranchuk
University of Colorado

92 PUBLICATIONS 659 CITATIONS

SEE PROFILE



Adam Storey
Auckland University of Technology

55 PUBLICATIONS 1,318 CITATIONS

SEE PROFILE



Andre Nelson
Victoria University Melbourne

42 PUBLICATIONS 639 CITATIONS

SEE PROFILE



John Cronin
Auckland University of Technology

447 PUBLICATIONS 21,067 CITATIONS

SEE PROFILE

REVIEW ARTICLE

Isometric training and long-term adaptations: Effects of muscle length, intensity, and intent: A systematic review

Dustin J. Oranchuk¹  | Adam G. Storey¹ | André R. Nelson² | John B. Cronin^{1,3}

¹Sports Performance Research Institute
New Zealand, Auckland University of
Technology, Auckland, New Zealand

²Institute for Health and Sport, Victoria
University, Melbourne, Victoria, Australia

³School of Health and Medical
Science, Edith Cowan University, Perth,
Western Australia, Australia

Correspondence

Dustin J. Oranchuk, Sports Performance
Research Institute New Zealand, Auckland
University of Technology, Auckland, New
Zealand.

Email: dustinoranchuk@gmail.com

Isometric training is used in the rehabilitation and physical preparation of athletes, special populations, and the general public. However, little consensus exists regarding training guidelines for a variety of desired outcomes. Understanding the adaptive response to specific loading parameters would be of benefit to practitioners. The objective of this systematic review, therefore, was to detail the medium- to long-term adaptations of different types of isometric training on morphological, neurological, and performance variables. Exploration of the relevant subject matter was performed through MEDLINE, PubMed, SPORTDiscus, and CINAHL databases. English, full-text, peer-reviewed journal articles and unpublished doctoral dissertations investigating medium- to long-term (≥ 3 weeks) adaptations to isometric training in humans were identified. These studies were evaluated further for methodological quality. Twenty-six research outputs were reviewed. Isometric training at longer muscle lengths (0.86%-1.69%/week, ES = 0.03-0.09/week) produced greater muscular hypertrophy when compared to equal volumes of shorter muscle length training (0.08%-0.83%/week, ES = -0.003 to 0.07/week). Ballistic intent resulted in greater neuromuscular activation (1.04%-10.5%/week, ES = 0.02-0.31/week vs 1.64%-5.53%/week, ES = 0.03-0.20/week) and rapid force production (1.2%-13.4%/week, ES = 0.05-0.61/week vs 1.01%-8.13%/week, ES = 0.06-0.22/week). Substantial improvements in muscular hypertrophy and maximal force production were reported regardless of training intensity. High-intensity ($\geq 70\%$) contractions are required for improving tendon structure and function. Additionally, long muscle length training results in greater transference to dynamic performance. Despite relatively few studies meeting the inclusion criteria, this review provides practitioners with insight into which isometric training variables (eg, joint angle, intensity, intent) to manipulate to achieve desired morphological and neuromuscular adaptations.

KEYWORDS

eccentric, fascicle, force, mechanical loading, muscle, stiffness, strength, tendon

1 | INTRODUCTION

Resistance training is widely utilized as a component of physical preparation for populations ranging from elite strength and power athletes to injured members of the general public.¹ Commonly documented resistance training adaptations

include increased muscle mass,² tendon quality,³⁻⁵ strength, power, and range of motion,⁶ delaying muscular fatigue,^{7,8} and improving voluntary activation.⁹ Dynamic movements incorporating the stretch-shortening cycle (SSC) comprise the overwhelming majority of resistance training programs.¹⁰ However, isolated concentric, eccentric, and isometric

contractions have specific advantages when improving musculo-skeletal properties and neuromuscular function¹¹⁻¹³ and are increasing in popularity.¹⁴ Isometric contractions (where the muscle-tendon unit remains at a constant length) and their role as a training option provide the focus of this paper.

Training with isometric contractions has been purported to have several advantages. First, isometric training allows for a tightly controlled application of force within pain-free joint angles in rehabilitative settings.^{15,16} Second, isometric training provides a means to induce force overload as maximal isometric force is greater than that of concentric contractions.¹⁷ Third, a practitioner who understands the physical demands of a sport may be able to utilize isometric training to focus on specific weak points in a range of motion that can positively transfer to performance¹⁸ and injury prevention.¹⁹ Isometric contractions can also be used to provide an acute analgesic effect and allow for pain-free dynamic loading^{20,21} by altering excitatory and inhibitory functions in the corticomotor pathways.²² Additionally, isometric contractions are a highly reliable means of assessing and tracking changes in force production.²³⁻²⁵ However, the ability of isometric assessments to predict dynamic performance is questionable,²³⁻²⁵ despite multi-joint appraisals showing promise.²⁶⁻²⁹

Isometric training can elicit changes in physiological qualities including muscle architecture,³⁰ tendon stiffness and health,^{21,31} joint angle-specific torque,³¹⁻³³ and metabolic functions.³⁴ As with any mode of resistance training, several variables can be manipulated to alter the stimulus. The most common isometric training variations include altering joint angles^{30-33,35-40} and contraction intensity or duration.^{34,39,41-47} Less frequently researched variations include contraction intent (eg, ramp vs ballistic)^{43,47,48} and incorporating special methods such as blood flow restriction,^{49,50} vibration,^{51,52} and electrical stimulation.⁵³ Additionally, emerging research has demonstrated unique neuromuscular characteristics between “pushing” (ie, exerting force against an immovable object) and “holding” (ie, maintaining a joint position while resisting an external force) isometric contractions.⁵⁴⁻⁶⁰ Understanding the loading parameters that achieve a desired adaptive response in muscle and tendon would be of benefit to practitioners. Therefore, the purpose of this review was to systematically evaluate research directly comparing the outcomes of isometric training variations and to provide training guidelines for a variety of desired outcomes.

2 | METHODS

The systematic review conformed to the “Preferred Reporting Items for Systematic Reviews and Meta-Analyses” (PRISMA) guidelines.⁶¹ Therefore, no Institutional Review Board approval was necessary.

2.1 | Literature search methodology

An electronic search was conducted utilizing MEDLINE, SPORTDiscus, PubMed, and CINAHL databases from inception to March 2018. Key terms were searched for within the article title, abstract, and keywords using conjunctions “OR” and “AND” with truncation “*.” Combinations of the following Boolean phrases comprised the search terms: (Isometric train*) AND (strength* OR stiff*); (Isometric train*) AND (muscle* OR tendon*); (Isometric train*) AND (session* OR week*).

2.2 | Inclusion and exclusion criteria

Studies were included in the review based on the following criteria: (a) full text available in English; (b) peer-reviewed journal publications or doctoral dissertations; and (c) the study compared two or more variations of isometric training. Studies were excluded if (a) they were conference papers/posters/presentations; (b) they focused on small joints or muscles such as fingers or toes; (c) primary dependent variables were related to cardiovascular health; (d) they included non-human subjects; (e) they were *in vitro*; (f) the intervention period was less than three weeks in duration; and (g) they included variables such as blood restriction, vibration, or electrical stimulation. Search strategy and inclusion/exclusion results are summarized in Figure 1.

2.3 | Quality assessment

Studies that met the inclusion criteria were assessed to determine their quality based on established scales utilized in the fields of sport and exercise science, kinesiology, health care, and rehabilitation. Adapted from a systematic review by Brughelli et al,⁶² the scale developed for the current review is illustrated in Data S1. Ten items were scored as 0 (clearly no), 1 (maybe), or 2 (clearly yes) based on this scoring rubric.⁶² Therefore, each study received a quality score ranging from 0 to 20. Two researchers completed the quality assessments of each paper with a third researcher settling any discrepancies in scoring.

2.4 | Statistical analysis

Percent change and Cohen’s *d* effect sizes (ES) were calculated wherever possible to indicate the magnitude of the practical effect. Effect sizes were averaged across the length of an intervention where applicable. As recommended by Rhea,⁶³ effect sizes were interpreted as follows: trivial <0.35, small = 0.35-0.80, medium = 0.80-1.50, and large > 1.5 for recreationally active participants.⁶³ Where possible, data were pooled and average ES change and % change (pre-post) per week were calculated. All reported ES and percentage changes are pre-post within-group, unless otherwise stated.

3 | RESULTS

A total of 26 studies with a mean quality score of 14.3/20 (range = 10-18) met the inclusion criteria for the review (Data S2). A total of 713 participants (463 male, 250 female) were recruited with an average sample size of 27.4 ± 28.1 (4-120). Of the accepted investigations, the mean age of the reported participants was 24.3 ± 3.3 years (19.3-31.8); seven studies failed to report participant mean age. Most studies (16/26) recruited untrained participants, while the remainder (11/26) utilized “active” or “recreationally trained” participants. None of the accepted studies examined competitive athletes or well-trained participants. All 26 accepted investigations clearly stated independent and dependent variables, and 10 included a non-exercise control group. The mean length of intervention was 8.4 ± 3.6 (range = 3-14) weeks, with an average of 3.5 ± 0.96 (range = 2-7) sessions per week for an average of 28.6 ± 13.2 (range = 15-56) total training sessions. Interventions were volume-equated in 17/26 studies, while 10/26 studies included a non-exercise control group. Closed-chain movements were only utilized in two studies, whereas 23/26 utilized single-joint contractions.

Nine published journal articles and one unpublished doctoral dissertation examining the chronic (5-12 weeks) effects of isometric training at varying joint angles fulfilled the inclusion criteria (Table 1).^{30-33,35-38,40,64} Of the ten included studies, eight centered on the knee extensors,^{30-33,35,38,40,64} with two utilizing the elbow flexors.^{36,37} Six published articles examining the effect of contraction intensity (Table 2)

fulfilled the inclusion criteria.^{41,42,44-46,65} Of these studies, three examined plantar flexors^{41,42,65} and one examined knee extensors,⁴⁶ while single studies examined the elbow flexors⁴⁵ and extensors, respectively.⁴⁴ Training variations outside of joint position or contraction intensity were also included. These variations include the following: (a) intent of contraction which included “progressive” vs “rapid”^{48,66} and “explosive” vs “sustained”^{43,47,67} contractions (Table 3); (b) total volume³⁹; (c) contraction duration^{13,34}; (d) rest period duration⁶⁸; and (e) periodization schemes⁶⁹ (Table 4).

When synthesizing statistically significant findings, measures of muscular size increased in nine studies (5%-19.7%, ES = 0.19-1.23) by 0.84%/week and 0.043 ES/week.^{13,30-32,34,43,44,67,69} Maximal isometric force significantly increased in 14 studies (8%-60.3%, ES = 0.34-3.26) by 4.34%/week and 0.20 ES/week.^{32,35,37,38,40,43,44,46-48,64-67} The comparison between joint angle and hypertrophic adaptation ($n = 3$ studies) revealed that training with joint angles $\leq 70^\circ$ ($46 \pm 6.9^\circ$) improved muscle size by an average of $0.47 \pm 0.48\%$ /week and 0.032 ± 0.037 ES/week, compared to $1.16 \pm 0.46\%$ /week and 0.067 ± 0.032 ES/week when training at $>70^\circ$ of flexion (Figure 2).³⁰⁻³² When comparing the nine studies that reported training joint angle and hypertrophic adaptations, training with joint angles $\leq 70^\circ$ ($59.8 \pm 11.1^\circ$) improved muscle size by an average of $0.61 \pm 0.42\%$ /week and 0.045 ± 0.034 ES/week, compared to $0.88 \pm 0.8\%$ /week and 0.046 ± 0.027 ES/week when training at $>70^\circ$ ($88.6 \pm 6^\circ$) of flexion (Data S3).^{13,30-32,34,43,44,67,69} The comparative effects of training intensity on muscular hypertrophy were that

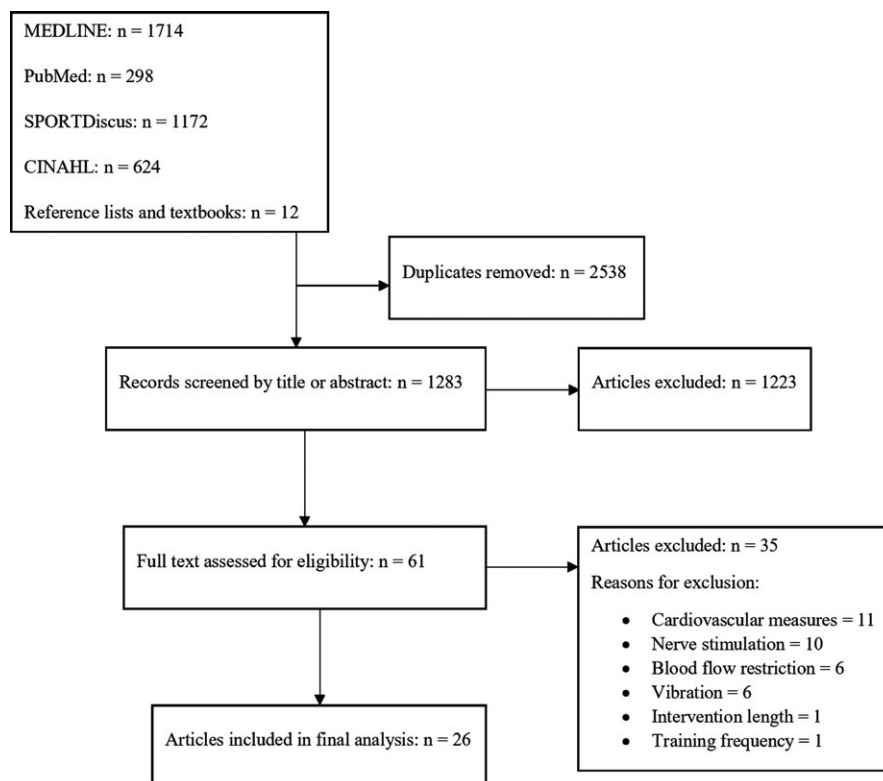


FIGURE 1 Search strategy

TABLE 1 Joint angle

Study, year (quality)	Subjects	Intervention	Mechanical and neural adaptations ($P < 0.05$, $ES \geq 0.50$)	Performance effect ($P < 0.05$, $ES \geq 0.50$)
Alegre, Ferri-Morales, Rodriguez-Casares, & Aguado (2014) ³⁰ (18/20)	Healthy, untrained university students M = 22 F = 7 19.3 years	Isometric knee extension SML = 50° LML = 90° ~74% of MVIC 8 wk, 2-3/wk	SML: ↑VL thickness at 25% and 50% muscle length (5.2%-6.1%, $ES = 0.23$ -0.24) ↑isokinetic EMG at 60-70° ($ES = 1.0$) and 50-60° ($P = 0.21$, $ES = 0.77$) LML: ↑VL thickness at 25%, 50%, and 75% muscle length (9%-13.5%, $ES = 0.31$ -0.65) ↑VL pennation angle (11.7%, $ES = 0.45$)	SML: ↓Optimum angle (7.3%, $ES = 0.91$) LML: ↑Concentric torque at 60° s ⁻¹ (22.6%, $ES = 1.1$) ↑Optimum angle (14.6%, $ES = 1.38$)
Bandy & Hanten (1993) ³⁸ (18/20)	Healthy, untrained, university students F = 107 23.9 y	Isometric knee extension SML = 30° MML = 60° LML = 90° 100% of MVIC 8 wk, 4/wk	SML: ↑EMG at 15, 30, 45 and 60° vs ↑EMG in control ($ES = 0.87$ -1.65) MML: ↑EMG at 15, 30, 45, 60 and 70° vs ↑EMG control ($ES = 0.36$ -2.26) LML: ↑EMG at 30, 45, 60, 75, 90, and 105° vs ↑EMG in control ($ES = 0.74$ -2.28)	SML: ↑MVIC at 15, 30, 45 and 60 ($ES = 0.88$ -1.94) MML: ↑MVIC at 15, 30, 45, 60 and 75° ($ES = 1.01$ -2.25) LML: ↑MVIC at 15, 30, 45, 60, 75, 90, and 105° ($ES = 0.94$ -3.26)
Bogdanis et al (2018) ⁶⁴ (15/20)	Healthy, active university students M = 15 21.5 ± 2.1 y	Isometric leg press (+countermove- ment jumps) SML = 35° of knee flexion LML = 95° of knee flexion 100% of MVIC 6 wk, 3/wk		SML: ↓Optimum angle (9.7%, $ES = 1.77$) ↑MVIC at 18° (22%, $ES = 0.88$) and 34° (57.4%, $ES = 2.41$) ↓RFD 0-200 ms and 0-300 ms at 80° (11.8%-13.8%, $ES = 0.51$ -0.60) ↑RFD 0-200 ms and 0-300 ms at 18° (40.7%-45.4%, $ES = 1.2$ -1.52) and 34° (17.9%-20.9%, $ES = 0.62$ -0.77) ↑1RM squat (9.6%, $ES = 0.61$) ↑CMJ height (7.2%, $ES = 0.66$) LML: ↑MVIC (main time effect: $P = 0.028$) at all joint angles (18-98°) (~12.3%) *↑RFD 0-300 ms at 34° (14.4%, $ES = 0.52$) ↑1RM squat (11.9%, $ES = 0.64$) ↑CMJ height (8.4%, $ES = 0.51$)
Kubo et al (2006) ³¹ (11/20)	Healthy university students M = 9 24 ± 1 y	Isometric knee extension SML = 50° LML = 100° 70% of MVIC 12 wk, 4/wk	SML: ↑Quadriceps muscle volume (10%, $ES = 0.82$) ↑EMG at all joint angles (3.1%-7.5%, $ES = 0.25$ -0.44) LML: ↑Quadriceps muscle volume (11%, $ES = 1.06$) ↑Tendon stiffness (50.86%, $ES = 1.22$) ↓Tendon elongation (-14.01%, $ES = 0.62$) ↑EMG at all joint angles (7%-8.84%, $ES = 0.45$ -0.72)	SML: ↑MVIC at 40, 50, 60, 70, and 80° LML: ↑MVIC at 40, 50, 60, 70, 80, 90, 100, and 110°

(Continues)

TABLE 1 (Continued)

Study, year (quality)	Subjects	Intervention	Mechanical and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$, ES ≥ 0.50)
Lindh (1979) ⁴⁰ (13/20)	Healthy F = 10 26.5 y	Isometric knee extension SML = 15° LML = 60° 100% of MVIC 5 wk, 3/wk		SML: ↑MVIC in SML at 15° (32%) ↑MVIC at 60° (14%) ↑Con torque at 30° s ⁻¹ LML: ↑MVIC at 15° (11%) ↑MVIC at 60° (31%) ↑Con torque at 30° s ⁻¹
Noorkoiv, Nosaka, & Blazevich (2014) ³² (17/20)	Healthy, untrained M = 16 23.7 ± 4.0 y	Isometric knee extension SML = 38.1 ± 3.7° LML = 87.5 ± 6.0° 100% of MVIC 6 wk, 3/wk	SML: ↑Mid-VL fascicle length (5.6%, ES = 0.63) LML: ↑Voluntary activation at 50° (ES = 0.53) and 60° (ES = 1.02) ↑Total quadriceps muscle volume (5.2%, ES = 0.19) ↑Distal VL fascicle length (5.8%, ES = 0.33)	SML: ↑MVIC at 40 and 50° (8.0%-14.2%, ES = 0.34-0.54)
Noorkoiv, Nosaka, & Blazevich (2015) ³³ (17/20)	Healthy, untrained M = 16 23.7 ± 4.0 y	Isometric knee extension SML = 38.1 ± 3.7° LML = 87.5 ± 6.0° 100% of MVIC 6 wk, 3/wk		LML: ↑Concentric torque at 30, *60, *90, and 120° s ⁻¹ (10.1%-13%, ES = 0.55-0.70)
Rasch & Pierson (1964) ³⁶ (13/20)	Healthy, untrained university students M = 29	Isometric elbow flexion Single-angle = 3 sets at 90° Multi-angle = 1 set at 60, 90 and 120° 100% of MVIC 5 wk, 5/wk		
Sterling (1969) ³⁵ (18/20)	University physical education students M = 120	Isometric "hip press" SML = 25° MML = 55° LML = 85° 100% MVIC 7 wk, 3/wk		SML: ↑MVIC at 25 and 55° (21%-37.2%) MML: ↑MVIC at 25 and 55° (15.4%-51.4%) LML: ↑MVIC at 85° (3.1%)
Thepaut-Mathieu, Van Hoecke, & Maton (1988) ³⁷ 11/20	Untrained M = 24 31.8 y	Isometric elbow flexion SML = 60° MML = 100° LML = 155° 80% MVIC 5 wk, 3/wk	SML, MML, and LML: ↑EMG at all angles	SML: ↑MVIC at 60 and 80° (10%-25%) MML: ↑MVIC at 60-155° (22%-30%) LML: ↑MVIC at 80-155° (24%-54%)

1RM, 1 repetition maximum; CMJ, countermovement jump; Con, concentric; ES, effect size (Cohen's *d*); LML, long muscle length; MML, medium muscle length; MVIC, maximal voluntary isometric contraction; RF, rectus femoris; SML, short muscle length; VL, vastus lateralis; VM, vastus medialis.

*Denotes $P > 0.05$.

intensities $\leq 70\%$ ($68.9 \pm 3.3\%$) of MVIC improved muscle size by $0.77 \pm 0.26\%$ /week and 0.13 ± 0.12 ES/week, compared to $0.70 \pm 0.55\%$ /week and 0.13 ± 0.21 ES/week when training at $>70\%$ ($85.3 \pm 12\%$) of MVIC (Figure 3).^{13,30-32,34,43,44,67,69} The comparisons of training intensity and improvements in isometric force ($n = 3$ studies) found that

training at $\leq 70\%$ ($41.3 \pm 16.5\%$) of MVIC improved muscle size by $6.8 \pm 3\%$ /week and 0.32 ± 0.13 ES/week, compared to $8.9 \pm 5.5\%$ /week and 0.36 ± 0.11 ES/week when training at $>70\%$ ($100 \pm 0\%$) of MVIC (Figure 4).^{44,46,65} The joint angle-isometric force comparison ($n = 7$) showed that training at $\leq 70^\circ$ ($42.8 \pm 16.4^\circ$) resulted in MVIC improvements

TABLE 2 Contraction intensity

Study, quality	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$, ES ≥ 0.50)
Adamantios Arampatzis, Karamanidis, & Albracht (2007) ⁴¹ 14/20	Healthy, untrained university students M = 7 F = 14 28 y	Isometric plantar flexion LI = 55% MVIC (24 contractions) HI = 90% MVIC (16 contractions) 14 wk, 4/wk	LI: ↑Tendon elongation (16.2%, ES = 0.56) ↑Tendon strain (17.4%, ES = 0.57) ↑Calculated maximum tendon force (28.4%, ES = 1.76) HI: ↑Tendon stiffness (36%, ES = 1.57) ↑Tendon CSA at 60% and 70% of tendon length ↑Calculated maximum tendon force (43.6%, ES = 2.04)	
Adamantios Arampatzis, Peper, Bierbaum, & Albracht (2010) ⁴² 14/20	Healthy, untrained university students M = 11 23.9 y	Isometric plantar flexion LI = 55% MVIC (20 contractions) HI = 90% MVIC (12 contractions) 14 wk, 4/wk	LI: ↑Tendon elongation (14%, ES = 0.84) ↑Tendon strain (13.7%, ES = 0.67) ↑Calculated maximum tendon force (11.7%, ES = 0.89) HI: ↑Tendon stiffness (17.1%, ES = 0.82) ↑Calculated maximum tendon force (11.9%, ES = 0.81)	
Kanehisa et al (2002) ⁴⁴ 16/20	Healthy, untrained M = 12 27.5 y	Isometric elbow extension LI = 60% MVIC (4 × 30 s) HI = 100% MVIC (12 × 6 s) 10 wk, 3/wk	LI: ↑Muscle volume (5.3%, ES = 0.26) HI: ↑Muscle volume (12.4%, ES = 0.28)	LI: ↑MVIC (61%, ES = 1.91) HI: ↑MVIC (60.3%, ES = 2.71)
Khouw & Herbert (1998) ⁴⁵ 11/20	51 untrained university students M = 18 F = 33	Isometric elbow flexion Each subject assigned to an individual intensity between 0% and 100% in 2% increments 6 weeks, 3/week	Greater ↑MVIC (slope = 0.19, 5.3%, $P = 0.006$) when training closer to 100%	
Szeto, Strauss, De Domenico, & Sun Lai (1989) ⁴⁶ 11/20	University students M = 6 F = 12	Isometric knee extension LI = 25% MVIC MI = 50% MVIC HI = 100% MVIC 3 wk, 5/wk	LI: *↑MVIC (22.3%, ES = 0.61) MI: ↑MVIC (31.3%, ES = 1.14) HI: ↑MVIC (45.7%, ES = 1.44)	

(Continues)

TABLE 2 (Continued)

Study, quality	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$, ES ≥ 0.50)
Young, McDonagh, & Davies (1985) ⁶⁵ 12/20	Healthy M = 4 20.5 y	Isometric plantar flexion LI = 30% MVIC (7-15 × 60 s) HI = 100% MVIC (3-s contractions) HI, 5 wk; and LI, 8 wk, 7/wk		LI: ↑MVIC (3.3%/wk) ↑MVIC (30.2%, ES = 2.22) ↑Fatigue index (19.4%, ES = 1.72) HI: ↑MVIC (5.5%/wk) ↑MVIC (21.2%, ES = 1.67)

ES, effect size (Cohen's d); HI, high intensity; LI, low intensity; MVIC, maximal voluntary isometric contraction.

*Denotes $P > 0.05$.

of $4 \pm 2.1\%$ /week and 0.15 ± 0.1 ES/week, compared to $3.4 \pm 4.2\%$ /week and 0.15 ± 0.17 ES/week when training at $>70^\circ$ ($101.8 \pm 24.2^\circ$) of flexion (Data S4).^{31,32,35,37,38,40,64}

4 | DISCUSSION

4.1 | Morphological adaptations

Adaptations to the physical structure of tissues can be caused by several factors, including mechanical, metabolic, and hormonal factors, and often result in altered function. The morphology of the musculo-skeletal system is of relevance to this review and provides the focus for subsequent discussion.

4.1.1 | Muscle volume

While most methods of progressive resistance training can result in increased muscular size, it is important to understand how to optimally alter variables including intensity, frequency, and duration of each training method for maximal efficiency. Isometric resistance training has been demonstrated to induce significant hypertrophy.^{13,30-32,34,39,43,44}

When comparing adaptations in muscle volume between isometric training variations, several patterns emerged, conforming to accepted dynamic training principles. Of the studies comparing isometric training at differing joint angles (Table 1), only three evaluated muscle volume or thickness.³⁰⁻³² All three studies found that isometric training at long muscle lengths (LMLs) was superior to equal volumes of training at short muscle lengths (SMLs) for increasing muscle size.³⁰⁻³² These findings are not surprising as a large portion of the existing literature has demonstrated that dynamic training through a large range of motion is beneficial when hypertrophy is desired.⁷⁰⁻⁷² Additionally, contractions at LML tend to produce higher quantities of muscle damage, likely by altering the joint moment arm and increasing mechanical tension when compared to a SML.⁷³ Contractions at LML also result in greater blood flow occlusion, rates of oxygen consumption, and metabolite buildup when compared to SML contractions.⁴⁹ These metabolic factors are well established to contribute to muscular hypertrophy.^{74,75}

While volume-equated isometric training leads to greater improvements in hypertrophy when performed at LMLs,^{30,32,33} the magnitude of hypertrophy was not significantly different in any of the seven included studies investigating/reporting training intensity.^{13,30-32,34,43,44} Interestingly, the pooled data of included study outcomes suggest that training intensity has a small effect on hypertrophy and explains little of the variation in hypertrophic adaptation (Figure 3). For example, Kubo et al¹³ compared the effects of load-equated isometric contractions held for short (~1 second) or long (20 seconds) periods of time. While both long- and short-duration

TABLE 3 Contraction intent

Study, year (quality)	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$, ES ≥ 0.50)
Balshaw, Massey, Maden-Wilkinson, Tillin, & Folland (2016) ⁴³ (15/20)	Healthy, untrained M = 43	Isometric knee extension MST = 1-s build to 75% of MVIC, hold for 3 s (40 contractions) EST = rapidly built to $\geq 80\%$ of MVIC and hold for 1 s (40 contractions) 12 wk, 3/wk	MST: $\uparrow$ Muscle volume (8.1%, ES = 0.50) $\uparrow$ EMG at MVIC (27.8%, ES = 0.67) $\uparrow$ EMG 0-150 ms (14.3%, ES = 0.36) EST: $\uparrow$ EMG 0-100 and 0-150 ms (12.5%-31.3%, ES = 0.26-0.67)	MST: $\uparrow$ MVIC (23.4%, ES = 1.19) $\uparrow$ Force at 150 ms (12.1%, ES = 0.74) EST: $\uparrow$ MVIC (17.2%, ES = 1.24) $\uparrow$ Force at 50, 100, and 150 ms (14.4%-32.6%, ES = 0.65-1.06)
Maffiuletti & Martin (2001) ⁴⁸ (17/20)	Healthy untrained M = 21	Isometric knee extension RC = 4 s to reach MVIC BC = 1 s to reach MVIC 7 wk, 3/wk	RC: $\downarrow$ VL EMG BC: $\uparrow$ Peak twitch (29.8%) $\downarrow$ Contraction time $\downarrow$ Maximal twitch relaxation	RC: $\uparrow$ MVIC at 55°, 65° (15.7%) and 75° $\uparrow$ Eccentric torque at 60° s ⁻¹ (15.6%) $\uparrow$ Concentric torque at 60 and 240° s ⁻¹ BC: $\uparrow$ MVIC at 55°, 65° (27.4%) and 75° $\uparrow$ Eccentric torque at 60° s ⁻¹ (18.3%) $\uparrow$ Concentric torque at 60 and 240° s ⁻¹
Massey, Balshaw, Maden-Wilkinson, Tillin, & Folland (2018) ⁶⁷ (18/20)	Healthy untrained M = 42 MST = 25 $\pm$ 2 y EST = 25 $\pm$ 2 y CON = 25 $\pm$ 3 y	Isometric knee extension MST = 1-s build to 75% of MVIC, hold for 3 s (~10 contractions) EST = rapidly built to ~80% of MVIC (~10 contractions) 12 wk, 3/wk	MST: $\uparrow$ Muscle volume (8.1%, ES = 0.47) $\uparrow$ VL aponeurosis area (5.9%, ES = 0.34) $\uparrow$ Tendon stiffness (14.3%, ES = 0.79) $\uparrow$ Young's modulus (14.4%, ES = 0.60) $\uparrow$ Tendon-aponeurosis stiffness (22.7%, ES = 0.54) EST: $\uparrow$ VL aponeurosis area (4.4%, ES = 0.38) $\downarrow$ Tendon CSA (2.8%, ES = 0.31) $\downarrow$ Tendon elongation (11%, ES = 0.75) $\uparrow$ Tendon stiffness (19.9%, ES = 0.95) $\downarrow$ Tendon strain (11.8%, ES = 0.56) $\uparrow$ Young's modulus (21.1%, ES = 1.13) $\uparrow$ Tendon-aponeurosis elongation (16%, ES = 1.0)	MST: $\uparrow$ MVIC (23.6%, ES = 1.17) EST: $\uparrow$ MVIC (16.7%, ES = 1.23)
Tillin & Folland (2014) ⁴⁷ (12/20)	Healthy, recreationally active male university students N = 19 MST = 20.9 $\pm$ 1.1 y EST = 20.2 $\pm$ 2.4 y	Isometric knee extension MST = 1-s build to 75% of MVIC, hold for 3 s (10 contractions) EST = rapidly built to $\geq 90\%$ of MVIC and hold for 1 s (10 contractions) 4 wk, 4/wk	MST: $\uparrow$ M-wave at MVIC (28.1%, ES = 1.28) $\downarrow$ %EMG at 50 and 150 ms (11.7%-22.1%, ES = 0.59-0.79) EST: $\uparrow$ M-wave at 50 and 100 ms (25%-42%, ES = 0.95-1.05)	MST: $\uparrow$ MVIC (20.5%, ES = 1.46) $\uparrow$ MVIC at 50, 100, and 150 ms (3.09%-7.39%, ES = 0.084-0.52) EST: $\uparrow$ MVIC (10.6%, ES = 0.56) $\uparrow$ MVIC at 50, 100, and 150 ms (13.1%-53.7%, ES = 0.96-1.2)

(Continues)

TABLE 3 (Continued)

Study, year (quality)	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$, ES ≥ 0.50)
Williams (2011) ⁶⁶ (15/20)	Healthy, untrained university students M = 11 F = 12 Ramp = 9 Ballistic = 8 22.8 y	Isometric knee extension RC = 4 s to reach MVIC BC = 1 s to reach MVIC 6 wk, 3/wk	RC: ↑Ramp VA (7.7%, ES = 1.99) ↑Ballistic VA (8.3%, ES = 1.75) *↑150 ms VA (9.82%, ES = 0.74) BC: ↑Ramp VA (4.1%, ES = 1.07) ↑Ballistic VA (7.9%, ES = 1.50) ↑150 ms VA (31.6%, ES = 1.84)	RC: ↑Ramp MVIC (20%, ES = 1.95) ↑Ballistic MVIC (17.8%, ES = 1.56) *↑150 ms force (14.3%, ES = 1.10) BC: ↑Ramp MVIC (15.7%, ES = 0.75) ↑Ballistic MVIC (18.9%, ES = 0.88) ↑150 ms force (48.8%, ES = 3.66)

BC, ballistic contraction; ES, effect size (Cohen's *d*); EST, explosive strength training; MST, maximal strength training; MVIC, maximal voluntary isometric contraction; RC, ramp contraction; VA, voluntary activation.

*Denotes $P > 0.05$.

contractions led to small, but significant increases in muscle thickness, there was little difference ($P > 0.05$) between groups (7.6%, ES = 0.38, $P = 0.023\%$ vs 7.4%, ES = 0.36, $P = 0.018$).¹³ Similarly, Kanehisa et al⁴⁴ employed ten weeks of volume-equated isometric training at either low (60%) or high (100%) intensity. While both low- and high-intensity training programs significantly increased triceps brachii hypertrophy, there was no statistical between-group difference ($P = 0.061$) in anatomical cross-sectional area (low: 12.1%, ES = 1.72 vs high: 17.1%, ES = 1.65).⁴⁴ However, high-intensity training had a greater effect on muscle volume than the lower intensity (12.4%, ES = 0.28% vs 5.3%, ES = 0.26; $P = 0.039$) despite nearly identical effect sizes.⁴⁴ These findings are in close agreement with recent studies and meta-analyses that concluded that hypertrophic adaptations are similar if total load is equated and training intensity is $>20\%$ of maximal voluntary contraction.^{76,77}

When the training volume is not equated between groups, it seems higher volumes are better for inducing muscular hypertrophy, regardless of contraction intensity. Meyers³⁹ compared low (3×6 seconds MVIC)- and high (20×6 seconds MVIC)-volume isometric training of the elbow flexors. Following the six-week intervention, the high-volume training program resulted in significantly greater improvements in muscle girth compared to the low-volume group ($P < 0.05$). Similarly, Balshaw et al⁴³ and Massey et al⁶⁷ compared “maximal strength” (40×3 seconds contractions, 75% of MVIC) and “explosive” (40×1 seconds contractions, 80% of MVIC) isometric training. Following the 12-week interventions, the “maximal strength” training groups experienced significant improvements in quadriceps muscle volume (8.1%, ES = 0.50, $P = 0.001$), whereas the “explosive” training groups (2.6%, ES = 0.17-0.26, $P = 0.195$ -0.247) did not.⁴³ Furthermore, the difference between groups was statistically significant ($P < 0.05$).^{43,67} Interestingly, Schott, McCully, and Rutherford³⁴ found that long-duration (4×30 second MVIC) contractions resulted in greater hypertrophic adaptations when compared to short ($4 \text{ sets} \times 10 \times 3$ second MVIC)-duration contractions despite total time-under-tension being equated between groups. Following 14 weeks, the long-duration contraction group significantly ($P = 0.022$) improved vastus lateralis anatomical cross-sectional area at the proximal (10.1%) and distal (11.1%) portions of the femur, whereas no significant hypertrophic adaptations were observed in the short-duration group ($P > 0.05$).³⁴ Schott, McCully, and Rutherford's³⁴ findings are somewhat surprising as both groups underwent the same time-under-tension. However, sustained contractions are known to restrict blood flow, reduce muscle oxygen saturation, and increase metabolite concentrations in the muscle^{78,79} stimulating hypertrophy via multiple local and systemic mechanisms.^{74,75} Additionally, muscle contractions at LML consume more

TABLE 4 Other independent variables

Study, quality	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$ and/or ES ≥ 0.50)
Kubo, Kanehisa, & Fukunaga (2001) ¹³ 14/20	Healthy, untrained M = 8 22.6 y	Isometric knee extension SC = 3 × 50 rapid contractions LC = 4 × 20 s 70% MVIC 12 wk, 4/wk	SC: ↑Muscle volume (7.4%, ES = 0.36) *↑Tendon stiffness (17.5%, ES = 0.57) ↑Elastic energy (25.6%, ES = 1.85) LC: ↑Muscle volume (7.6%, ES = 0.38) ↑Tendon stiffness (57.3%, ES = 1.38) ↑Elastic energy (12%, ES = 0.58)	SC: ↑MVIC (49%, ES = 2.47) LC: ↑MVIC (41.6%, ES = 2.21)
Meyers (1967) ³⁹ 13/20	Healthy university students M = 29	Isometric elbow flexion LV = 3 × 6 s HV = 20 × 6 s 100% MVIC 6 wk, 3/wk	LV: ↑Muscle girth at 170° in trained arm HV: ↑Muscle girth at 170° in trained and untrained arm ↑Muscle girth at 90° in trained arm	LV: ↑MVIC at 170° (15.4%, ES = 0.93) *↑Muscle endurance (49.7%, ES = 0.71) HV: ↑MVIC at 170° (15.5%, ES = 0.46) *↑MVIC at 90° (9%, ES = 0.50) ↑Muscle endurance (42.7%, ES = 0.67)
Schott, McCully, & Rutherford (1995) ³⁴ 10/20	Healthy, untrained M = 1 F = 6 22.7 y	Isometric knee extension SC = 4 × 10 × 3 s LC = 4 × 30 s 70% of MVIC 14 wk, 3/wk	LC: ↑Muscle ACSA at lower (11.1%) and upper (10.1%) femur	SC: ↑MVIC (31.5%) ↑Concentric torque at 120 and 180° s ⁻¹ (11.3%-11.6%) LC: ↑MVIC at 90° (54.7%)
Ullrich, Holzinger, Soleimani, Pelzer, Stening, & Pfeiffer (2015) ⁶⁹ 16/20	Healthy, active university students F = 10 24.4 ± 3.2 y	Isometric knee extension TP limb = 3 wk 60%, 4 wk 80%, 3 wk 60%, 2 wk 80% of MVIC DUP limb = Alternating sessions at 60% and 80% of MVIC in one limb 16 wk, 2/wk	TP: ↑Thigh circumference (6.2%, ES = 0.45) ↑VL thickness at 25%, 50%, and 75% muscle length (15.5%-18.5%, ES = 0.98-1.23) ↑VL fascicle length (13.7%, ES = 1.17) ↑MVIC EMG (45%) DUP: ↑Thigh circumference (5.0%, ES = 0.37) ↑VL thickness at 25%, 50%, and 75% muscle length (12.4%-19.7%, ES = 0.72-1.01) ↑VL fascicle length (14.2%, ES = 0.90) ↑MVIC EMG (46%)	TP: ↑MVIC (24%) ↑Concentric torque at 60° s ⁻¹ (19%) DUP: ↑MVIC (23%) ↑Concentric torque at 60° s ⁻¹ (15%)

(Continues)

TABLE 4 (Continued)

Study, quality	Subjects	Intervention	Morphological and neural adaptations ($P < 0.05$, ES ≥ 0.50)	Performance effect ($P < 0.05$ and/or ES ≥ 0.50)
Waugh, Alktebi, De Sa, & Scott (2018) ⁶⁸ 14/20	Healthy, physically active M = 8 F = 10 30.1 ± 7.9 y	Isometric plantar flexion SR = 3 s between reps LR = 10 s between reps 90% MVIC 12 wk, 3/wk	SR: ↑Echo-type II (collagen reorganization) SR & LR: ↑Stiffness ↑Tendon stress ↑Young's modulus ↓Strain % ↓Tendon elongation	SR & LR: ↑MVIC

DUP, daily undulating periodization; ES, effect size (Cohen's d); HV, high volume; LC, long contraction; LR, long rest; LV, low volume; MVIC, maximal voluntary isometric contraction; SC, short contraction; SR, short rest; TP, traditional periodization.

^aDenotes $P > 0.05$.

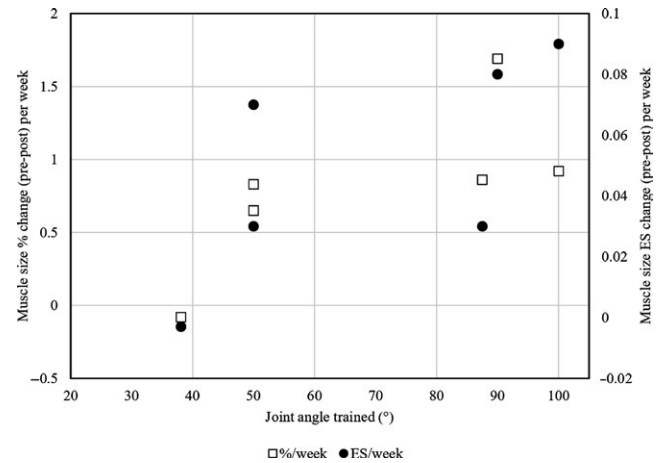


FIGURE 2 Isometrically trained joint angle and hypertrophic adaptations (N = 3)

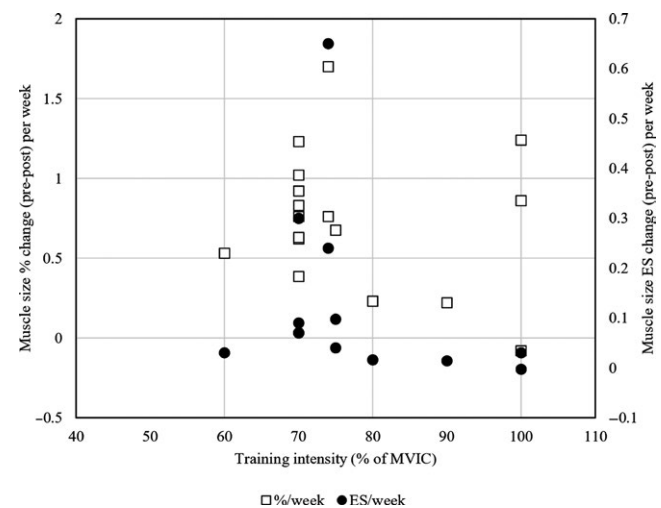


FIGURE 3 Isometric training intensity and hypertrophic adaptations (multiple comparison, N = 9)

oxygen,⁴⁹ which may in part explain the advantage of LML training when muscular hypertrophy is the primary goal.

4.1.2 | Muscle architecture

Unlike muscle volume, which is highly dependent on total training volume, there are demonstrable differences between contraction type and alteration in fascicle length and pennation angle.⁸⁰ To date, very few studies have compared the effect of isometric resistance training variations on muscle architecture; of those that have, results are equivocal. Noorkoiv, Nosaka, and Blazevich³² compared isometric training at SML ($38.1 \pm 3.7^\circ$ knee flexion) and LML ($87.5 \pm 6^\circ$ knee flexion). Interestingly, the vastus lateralis fascicle length at the mid-portion of the femur significantly increased following SML (5.6%, ES = 0.63, $P = 0.01$), but not LML (3.8%, ES = 0.34,

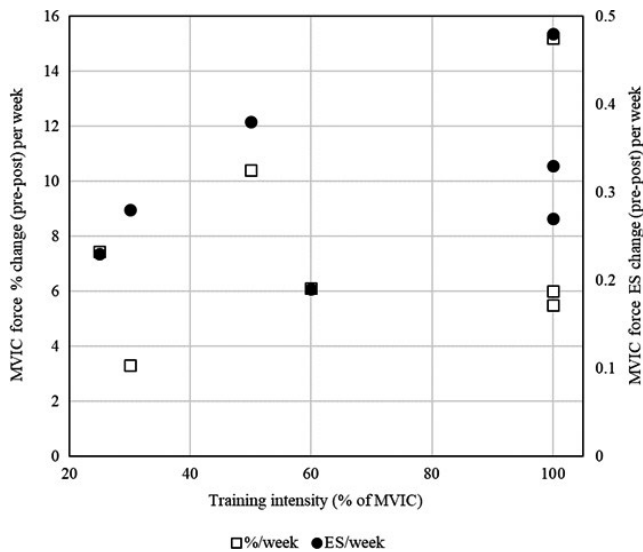


FIGURE 4 Isometric training intensity and force production (N = 3)

$P = 0.20$) training.³² Conversely, LML (5.8%, ES = 0.33, $P = 0.02$) significantly ($P = 0.01$) outperformed SML training (−1.1%, ES = 0.04, $P > 0.05$) for increasing distal fascicle length of the same muscle.³² Furthermore, LML training resulted in greater ($P < 0.01$) physiological cross-sectional areas in three of the four quadriceps heads, whereas the SML training did not ($P > 0.05$).³² Only one other isometric training comparison study reported meaningful shifts in muscle architecture and found that vastus lateralis pennation angle increased following LML (10.6%, ES = 0.45, $P = 0.038$), but not SML training (6.5%, ES = 0.46, $P = 0.076$).³⁰ However, Alegre et al³⁰ only measured the vastus lateralis pennation angle at the midpoint of the femur and potentially missed out on possible adaptations at the distal portion of the muscle.

4.1.3 | Tendon morphology

The primary function of the tendon is to transfer forces between bone and muscle, facilitating joint motion.⁵ Although originally assumed to be inert, tendinous structures can experience adaptations and are capable of significant architectural adaptations from habitual loading and injury.^{3-5,81-83} Injured tendons tend to be less stiff, despite increased thickness⁸⁴ due to a shift in viscoelastic properties.⁵ Additionally, tendinopathy negatively affects tendon structure, leading to increased vascularization and overall thickness.^{5,84} Although long-term alteration in tendon morphology is minimal in healthy, mature human tissue,⁵ tendons can increase in stiffness to optimize the time and magnitude of force transmission between muscle and bone.^{3,4,82} Conversely, healthy increases in tendon thickness and stiffness in response to exercise have been found to be region specific and may have rehabilitative, pre-habilitative, and performance benefits.^{3,4,20,81,82} For

instance, heavy (resistance) training can lead to an increase in maximal muscular force and rate of force development by increasing tendon stiffness, thus reducing the electromechanical delay.^{5,83,85} Additionally, increased tendon stiffness through chronic loading can be due to increased tendon CSA without alterations in viscoelastic properties, potentially improving safety when performing ballistic movements.⁵ While widely used in rehabilitation settings, there is a general lack of information regarding what isometric training variables are important for triggering specific tendinous adaptations.

Of the studies included in this review, only six directly assessed tendon structure or function. Two studies compared contraction intensity,^{41,42} with others examining the effects of contraction length,¹³ intent,⁶⁷ rest periods,⁶⁸ and joint angle.³¹ Arampatzis et al^{41,42} compared 14-week training programs consisting of volume-equated isometric plantar flexion at low (~55%) or high (~90%) intensities. Both investigations found increased Achilles tendon CSA and stiffness following high-intensity (17.1%–36%, ES = 0.82–1.57, $P < 0.05$), but not low-intensity (−5.2% to 7.9%, ES = 0.26–0.37, $P > 0.05$) training.^{41,42} Furthermore, tendon elongation under stress (an indication of elasticity) increased following low-intensity (14.0%–16.1%, ES = 0.56–0.84, $P > 0.05$), but not high-intensity (−1.4% to 3.9%, ES = 0.06–0.20, $P > 0.05$) training.^{41,42} Additionally, the included studies only compared isometric training at ~55 and 90% of MVIC which leaves a large range of potential intensities. However, previous interventions have reported large increases (17.5%–61.6%, ES = 0.57–4.9, $P < 0.05$) in tendon stiffness following training between 70% and 100% of MVIC.^{11,13,85} Therefore, it might be that a minimum intensity of ~70% MVIC is required to induce meaningful changes in tendon thickness and stiffness.

While only a single study has examined the effect of isometric training at different muscle lengths on tendon adaptation,³¹ the results tend to support a paradigm of LML training being superior to SML training. Kubo et al³¹ trained the knee extensors at either 50° or 90° of flexion and observed a significantly greater increase in tendon stiffness ($P = 0.021$) following LML (50.9%, ES = 1.22, $P = 0.014$), when compared to SML training (6.7%, ES = 0.26, $P = 0.181$). Similarly, distal tendon and deep aponeurosis elongation decreased following LML training (−14%, ES = 0.62, $P = 0.034$), whereas the SML group experienced a trivial increase (3.9%, ES = 0.15, $P > 0.05$). When comparing isometric contraction duration and tendon adaptations, only a single study exists.¹³ While both long (57.3%, ES = 1.38, $P = 0.003$) and short (17.5%, ES = 0.57, $P = 0.217$) contraction durations increased tendon stiffness, a significant between-group difference was reported ($P = 0.045$).¹³ Additionally, no significant differences in tendon elongation were present in either long (−2.2%, ES = 0.19, $P > 0.05$)- or short (4.1%, ES = 0.29, $P > 0.05$)-contraction-duration groups. Similarly, calculated elastic energy absorption increased in both long (12%, ES = 0.58, $P = 0.007$)- and

short (25.7%, $ES = 1.85$, $P = 0.002$)-duration groups with no significant difference between groups ($P = 0.056$) despite large differences in percent change and effect sizes along with a relatively low P -value. While the total time-under-tension was equalized between groups, the one-second duration of the short contraction group meant that a larger relative proportion of each effort would be spent building isometric force. Therefore, the maximal-force time-under-tension was not equalized.¹³ Similar to muscle tissue, tendon adaptations are responsive to chronic changes in total mechanical load^{3,86,87}; therefore, the potentially greater load in the long contraction group could explain the discrepancy in tendinous adaptations.

Massey et al⁶⁷ were the only researchers comparing contraction intent on morphological tendon adaptations. Both “maximal strength training” and “explosive strength training” produced significant improvements in vastus lateralis aponeurosis area (5.9%, $ES = 0.34\%$ vs 4.4%, $ES = 0.38$), Young’s modulus (14.4%, $ES = 0.60\%$ vs 21.1%, $ES = 1.13$), and tendon stiffness (14.3%, $ES = 0.79\%$ vs 19.9%, $ES = 0.95$).⁶⁷ However, only the “explosive strength training” group experienced significant increases in tendon-aponeurosis complex elongation (16%, $ES = 1.0$ vs -2.96 , $ES = 0.10$) and decreased tendon CSA (-2.8% , $ES = 0.31\%$ vs 0.41% , $ES = 0.03$), tendon elongation (-11% , $ES = 0.75\%$ vs -4.95% , $ES = 0.27$), and tendon strain (-11.8% , $ES = 0.56$ vs -4.17 , $ES = 0.19$).⁶⁷ Therefore, intent and rate of contraction appear to be an important training consideration. Lastly, Waugh et al⁶⁸ compared load-equated isometric plantar flexions with intra-contraction rest periods of 3 or 10 seconds. While there were differences ($P > 0.05$) in type I and type II collagen (factors in fiber reorganization),^{88,89} there were no between-group discrepancies ($P > 0.05$) in any other dependent variables following the 14-week intervention.⁶⁸ These data support a paradigm of a threshold intensity for mechanical loading to achieve tendon adaptations.^{86,87}

4.2 | Neurological adaptations

Of the 23 studies included in this review, 12 directly measured neural function.^{13,30-32,37,38,43,47,48,65,66,68} Of these 12 studies, it is notable that one did not report any neurological data in their results,⁶⁸ while two reported no significant changes following training, regardless of the condition.^{13,65} When examining electromyography (EMG) amplitude assessed through EMG, a clear trend existed between the studies comparing isometric training at different muscle lengths. Electromyographic amplitude tends to increase by larger magnitudes and over a larger range of joint angles following LML training, compared to training at SML. For example, Bandy and Hanten³⁸ examined isometric knee extension training at SML (30°), medium muscle length (MML; 60°), and LML (90°), assessing EMG amplitude at seven joint angles from 15° to 105° of flexion. Medium-to-large ($ES = 0.74$ -2.28) improvements

at six joint angles were observed following LML training, whereas MML and SML training only improved EMG activity at five ($ES = 0.36$ -2.26), and four ($ES = 0.87$ -1.65) of the assessed joint angles, respectively.³⁸ Similarly, Kubo et al³¹ observed larger increases in EMG activity at all measured angles following LML (7%-8.8%, $ES = 0.45$ -0.72) compared to SML (3.1%-7.5%, $ES = 0.25$ -0.44) training. Conversely, Alegre et al³⁰ reported an increase in EMG amplitude in favor of the SML training group, the only investigation to do so. Although the magnitude of increases in EMG amplitude was medium-large, the changes were limited to 50-60° ($ES = 0.77$, $P = 0.205$) and 60-70° ($ES = 1.0$, $P = 0.36$) of knee flexion during isokinetic knee extensions.³⁰ These findings are consistent with the findings of other investigations in that alterations in EMG amplitude are most specific at shorter muscle lengths.^{37,71,72}

All four studies comparing the effects of isometric training with different contraction intents (ballistic vs ramp) assessed neurological and neuromuscular adaptations via EMG and peripheral nerve stimulation interpolated twitch.^{43,47,48,66} As expected, adaptations were specific to the intent utilized in training. For example, Balshaw et al⁴³ examined the effects of 12 weeks of “maximal strength training” (1-second build to ~75% of MVIC and maintain for 3 seconds), with “explosive strength training” (rapid build to $\geq 90\%$ of MVIC and maintain for 1 second). The improvements in EMG amplitude at MVIC were larger ($ES = 0.36$, $P = 0.370$) following “maximal strength training” (27.8%, $ES = 0.67$, $P < 0.001$) compared to “explosive strength training” (19.1%, $ES = 0.44$, $P = 0.099$). Conversely, “explosive strength training” (31.3%, $ES = 0.67$, $P = 0.003$) increased EMG activity to a greater ($P < 0.001$) degree during the 0- to 100-ms and 0- to 150-ms period of muscle contraction compared to “maximal strength training” (14.3%, $ES = 0.36$, $P = 0.009$).⁴³ Additionally, only the rapid contraction group significantly increased EMG amplitude in the first 100 ms of muscle contraction (12.5%, $ES = 0.26$, $P = 0.048$).⁴³ Similarly, previous investigations examining contraction intent found greater improvements in EMG amplitude during MVIC with MST (1.28%-7%/week, $ES = 0.06$ -0.33/week) when compared to EST (0.68%-1.31%/week, $ES = 0.18$ -0.25/week).^{47,48,66} Furthermore, participants training with a ballistic intent (1.04%-10.5%/week, $ES = 0.26$ -0.31/week) achieved greater improvement in EMG amplitude during the initial 150 ms of maximal contraction when compared to MST (2.93%-5.53%/week, $ES = 0.03$ -0.07/week).^{43,47,48,66} These findings support the principle of training specificity as only the groups who intended to produce force quickly improved in that regard.

4.3 | Performance enhancement

Isometric training is commonly prescribed in rehabilitation settings, or early in physical preparation plans as a means

to increase neuromuscular, musculo-skeletal, and proprioceptive function. It is thought that the aforementioned improvements will later transfer to dynamic performance once specific movement patterns are integrated into the physical preparation plan. Despite existing literature reporting benefits of isometric training on multi-joint dynamic performance,^{11,85,90} none of the studies included in the current review included dynamic multi-joint assessments.

4.3.1 | Isometric peak force

Only four studies included in the present review directly compared MVIC production between group training at different intensities.^{44-46,65} Isometric peak force is considered a highly reliable measure, with a growing body of research reporting the validity of isometric assessments for assessing health and athletic performance.^{28,91} While training specificity is a major factor in performance improvements, if MVIC force is the desired outcome there does not appear to be a clear advantage to training at high or low intensities (Figure 4). Szeto et al⁴⁶ was the only study that reported statistically significant improvements in MVIC force in some, but not all training groups. Szeto et al⁴⁶ had subjects train their knee extensors at 25%, 50%, or 100% of MVIC. Following 15 sessions over three weeks, the group training at 25% did not experience statistically significant strength improvements despite medium effect sizes (22.3%, ES = 0.61, $P = 0.085$).⁴⁶ Conversely, large and statistically significant improvements were observed when training at 50% (31.3%, ES = 1.14, $P = 0.002$) and 100% (45.7%, ES = 1.44, $P = 0.013$) of MVIC.⁴⁶ However, time-under-tension, not total load, was equalized between groups, meaning that the 50% training group produced twice as much total force as the 25% group. While no data about fatigue are presented, it could be hypothesized that the group training with maximal effort underwent significantly greater loading than the other groups.⁴⁶ Additionally, the inclusion of a perceived effort or fatigue scale may have been valuable.

A clear pattern can be observed when comparing maximal force production following training at different muscle lengths. Despite LML resulting in greater hypertrophic adaptations, there is no difference in maximal force production at the trained joint angle between SML and LML interventions when analyzing the seven studies that directly compared joint angles (Data S4).^{31,32,35,37,38,40,64} However, transfer to non-trained joint angles is much lower following SML training. For example, Bandy and Hanten,³⁸ Bogdanis et al,⁶⁴ Kubo et al,³¹ and Thepaut-Mathieu, van Hoecke, and Maton³⁷ all trained participants at different muscle lengths and measured MVIC at numerous joint angles pre- and post-training. Bandy and Hanten³⁸ observed significant ($P < 0.05$) improvements at four, five, and seven of the tested joint angles following SML, MML, and LML, respectively. Bogdanis et al⁶⁴ reported increased MVIC at

two of the assessed joint angles following SML training (22%-57.4%, ES = 0.88-2.41), while the LML group improved in all six angles (~12.3%). Similarly, the SML group in Kubo et al's³¹ investigation significantly ($P < 0.05$) improved MVIC at five angles, while the LML group experienced significantly improved force production at eight of the tested angles. Interestingly, Thepaut-Mathieu, Van Hoecke, and Maton³⁷ found that their LML group significantly ($P < 0.05$) improved at four angles, compared to two and five angles in the SML and the MML group, respectively. These data suggest that LML and MML isometric resistance training is superior to SMLs when the aim is to improve force throughout a range of motion.

4.3.2 | Length-tension

The length-tension relationship, typically assessed by isometric or isokinetic contractions, is defined as the muscle length or joint angle at which peak force/torque is produced.⁹² Many studies have demonstrated acute optimal angle/length shifts toward longer muscle lengths following concentric, isometric, and eccentric exercise.^{73,93-98} Additionally, eccentric resistance training and training over a larger range of motion are well established for increasing the optimal angle long-term.^{70,95} It is plausible that the same relationship exists between muscle length and a shift in the optimal angle following isometric contractions. However, only a single study included in this review reported the angle of peak isokinetic torque,³⁰ while another examined optimal angle through an isometric leg press.⁶⁴ Alegre et al³⁰ observed a shift of 11° (14.6%, ES = 1.1, $P = 0.002$) toward longer muscle lengths following eight weeks of training at LML, whereas the SML group experienced a shift of 5.3° (7.3%, ES = 0.91, $P = 0.039$) in the opposite direction. Likewise, Bogdanis et al⁶⁴ reported a decrease in optimal angle following SML training (−9.7%, ES = 1.77) while the optimal angle was maintained in the LML group. While length-tension curve shifted toward the angle of training in several other studies, none were significant or altered the angle at which maximal isometric force was produced.³⁰ While a very limited sample, the report of Alegre et al³⁰ is unsurprising given that isometric exercise at LMLs is preferable to SMLs for acutely altering the length-tension relationship.⁹⁹ Finally, it should be noted that no included study reported any significant differences in isometric or isokinetic length-tension curves between group training with different intensities, contraction intents, or any other independent variable.

4.3.3 | The rate of force development

The rate of force development (RFD) is an important measurement in sports performance, as force application in many activities occurs over short time periods.^{14,100-102} Therefore,

while peak force is a valid and highly reliable means of broadly monitoring neuromuscular function, rapid force production characteristics are equally valuable and more specific to the execution of explosive tasks.^{2,100-103} Unfortunately, only three training studies examining different contraction intents reported RFD variables.^{43,47,48} Regardless, all three studies reported that isometric training with an “explosive” or “ballistic” intent was superior to ramping contractions for improving rapid force production.^{43,47,66} These findings align with the previously discussed alterations in EMG amplitude between contraction intents. For example, Williams⁶⁶ compared the adaptations following ballistic or ramp isometric training. While the ramp group experienced larger improvements in MVIC (ramp, 17.8%-20%, ES = 1.56-1.95, $P = 0.0008$ vs ballistic, 15.7%-18.9%, ES = 0.75-0.88, $P = 0.0036$), only the ballistic training group significantly improved voluntary activation (31.6%, ES = 1.84, $P = 0.0096$) and force at 150 ms (48.8%, ES = 1.29, $P = 0.0074$).⁶⁶ Similar findings are reported by Balshaw et al⁴³ and Tillin and Folland⁴⁷ where only the ballistic training groups significantly ($P < 0.05$) improved force at 50 and 100 ms (Table 3). These findings are not surprising, as several researchers have reported increased rapid force and power production, driven heavily by neurological alterations.¹⁰⁴⁻¹⁰⁶ Additionally, there is evidence to suggest that the intent of movement may be of similar value to actual external contraction velocity when improving RFD characteristics.¹⁰⁷

4.3.4 | Dynamic performance

The transferability of isometric resistance training to dynamic performance is questionable, despite specific isometric assessments closely relating to sports performance.⁹¹ Likewise, the degree of transference of isokinetic contraction to real-world movements has yet to be elucidated fully.^{24,26,27} Regardless, isokinetic testing provides a valuable means of assessing dynamic performance. Five studies utilized isokinetic assessments with three comparing various trained joint angles^{30,40,48} and two studies comparing contraction intent⁴⁸ or length of contraction, respectively.³⁴ Maffiuletti and Martin⁴⁸ reported similar improvements in eccentric torque at 60° s^{-1} and concentric torque at slow (60° s^{-1}) and faster (120° s^{-1}) angular velocities regardless of contraction intent. When comparing isometric training at different muscle lengths, Alegre et al³⁰ and Noorkoiv et al³³ observed significant ($P < 0.05$) improvements after training at LML, but not SML in concentric torque at 60 and 30° s^{-1} , 60, 90, and 120° s^{-1} , respectively, despite no significant differences in MVIC improvements between groups. Conversely, Lindh⁴⁰ reported that neither SML nor LML training groups improved isokinetic torque at 180° s^{-1} while both groups significantly ($P < 0.01$) improved peak torque at 30° s^{-1} . Finally, Bogdanis et al⁶⁴ observed similar improvements in one repetition maximum squat (9.6%, ES = 0.61% vs 11.9%, ES = 0.64) and

countermovement jump height (7.2%, ES = 0.66% vs 8.4%, ES = 0.51) following SML and LML leg press training, respectively. One possible explanation for these findings is that the LML training groups in Alegre et al³⁰ and Noorkoiv et al³³ experienced larger hypertrophic adaptations than the corresponding SML participants. Unfortunately, neither Lindh⁴⁰ nor Bogdanis et al⁶⁴ assessed morphological adaptations, making further analysis difficult.

4.4 | Applications

While the direct transfer of isometric resistance training to dynamic movements is questionable, physiological adaptations such as increased muscle mass and improved tendon qualities are beneficial in a variety of contexts. There is a well-established relationship between muscle mass, strength, and functional performance in a variety of activities and populations.¹⁰⁸⁻¹¹⁰ While it may require specific training in a movement to optimize neuromuscular performance,^{71,111} it is clear that producing and maintaining muscle mass and strength should be a priority for athletes and special populations alike. For this reason, isometric contractions are regularly used in rehabilitation programs and during specific training phases where dynamic contractions may be contraindicated.

The long-held belief that isometric resistance training should occur at the most important angle present in a dynamic activity holds true¹¹²⁻¹¹⁵ as the largest improvements in neuromuscular function occur at the trained angle.^{31,32,37,38,40} However, large neurological discrepancies exist between isometric and dynamic movements²⁵ suggesting that static training may not be an effective strategy for directly improving sports performance and should be primarily employed to alter morphology. Therefore, isometric training should occur predominantly at relatively LMLs as there is a clear advantage for improving muscle volumes (Figure 2) and strength throughout a range of motion.^{30-33,37,38} Additionally, large increases in tendon stiffness following LML have been reported, which would likely reduce electromechanical delay and therefore improve RFD.^{5,31,116} Furthermore, LML isometric training may have beneficial effects on the length-tension relationship,³⁰ although greater evidence is needed to solidify optimal angle as a key variable in performance and injury prevention.⁹² Similarly, architectural qualities of muscle may underpin the length-tension relationships. However, Alegre et al³⁰ observed no significant ($P > 0.05$) shift in fascicle length regardless of training angle, while Noorkoiv et al³² reported conflicting findings depending on which quadriceps head was evaluated. Therefore, isometric resistance training, regardless of muscle length, appears unlikely to efficiently lengthen muscle fascicles.

Training intensity is a key variable prescribed in intelligently designed resistance training programs. Evidence suggests that high-intensity resistance training is superior for improving force

production.^{45,76,117} However, the studies cited in this review show a questionable relationship between intensity and force production adaptations (Figure 4).^{13,30-32,34,43,44,46,65} Consistent with recent original research and meta-analyses, isometric training intensity does not appear to affect hypertrophic adaptations.^{76,77} While the lack of relationship between contraction intensity and force production is somewhat surprising, previous literature has reported that submaximal intensities can produce similar strength improvements when taken to failure, or when the volume is equated between groups.^{77,118} These findings suggest that isometric training intensity is not important when aiming to improve force production or alter muscle morphology. Therefore, increasing contraction durations,³⁴ increasing total volume, or shifting to longer muscle lengths^{30-32,38,40} is likely more efficient means of progressing isometric resistance training if strength and muscle size are a priority. Conversely, high-intensity ($\geq 70\%$ of MVIC) isometric contraction exclusively produced increased tendon thickness and stiffness.^{41,42} As overly compliant tendons are often an issue in untrained and injured populations, progressively increasing intensity during isometric contractions may be a safe and efficient means of preparing tendinous tissue for future dynamic loading.^{12,82} Additionally, sports requiring a high degree of reactive strength require relatively stiff tendinous structures to optimize performance.^{90,119,120}

Isometric training, like other modes of resistance exercise, should be executed in a way that most closely relates to the primary outcome goal. When muscular hypertrophy or maximal force production is the priority, the evidence demonstrates that there is little difference between contractions completed with a ballistic or a gradual ramp to the prescribed intensity.^{43,47,48,66} However, if rapid force production takes precedence, as it would in several sports, then isometric contractions should be performed as such.^{43,47,66} Conversely, ballistic contractions may be contraindicated or cause excessive pain in rehabilitative or special populations,²⁰ despite potential to provide unique morphological tendon adaptations.⁶⁷ Therefore, while ballistic contractions offer unique neuromuscular benefits, sustained contractions generally offer similar or greater morphological adaptations that are likely of interest to a wider variety of trainee.^{43,48,66}

4.5 | Limitations and directions for future research

While trends, or lack thereof, are evident in many of the key independent variables discussed in the current review, several limitations exist. While the widely homogeneous populations inter- and intra-study allowed for simple analysis, none of the included studies utilized special populations such as patients with tendon disorders, high-performance athletes, or experienced resistance trainees. Researchers and practitioners alike need to be cognizant of this limitation if wishing to generalize findings. Similarly, very few of the included

studies examined the effect of isometric training on dynamic performance, and only one utilized closed-chain or functional performance tasks in their testing batteries. Finally, while 26 studies were included, the large variety of independent and dependent variables made extensive inter-study analysis difficult and hence definitive conclusions problematic.

While the limitations present are broad, several directions for interesting future research exist. Isometric resistance training is often utilized by strength and conditioning coaches early in a training plan with the intent of preparing muscle and tendon morphologies for future dynamic loading. However, to the authors' knowledge, no published studies have examined the effect of a proceeding isometric training phase on dynamic or ballistic training periods despite a rise in popularity with this approach.¹⁴ On a related note, a limited number of studies have examined isometric training with free-weights. Isometric contraction intensity does not play a large role in driving morphological or neuromuscular adaptations, and total volume is likely a more important variable. However, resistance training modes have specific load cutoff points for altering tissue or neural properties.^{1,10} As such, future studies should aim to establish approximate weekly loading guidelines for a variety of populations, muscle groups, and dependent variables. Another interesting direction is determining whether isometric training can improve dynamic muscular endurance. Unfortunately, only a single included study evaluated fatigue,⁶⁵ and no studies examined fatigue during dynamic or stretch-shortening cycle activities such as cycling or running.

Another avenue for research geared toward rehabilitative populations is a multivariate examination of contraction intensity and joint angles. Physical therapists often prescribe isometric training as a means to stimulate morphological adaptations and improve neuromuscular function while tightly maintaining a pain-free range of motion. Anecdotally, therapists often limit isometric contractions to moderate joint angles as the increased ligament strain and pressure synonymous with maximal contraction intensities at large degrees of joint flexion may cause unwanted pain and inhibition.^{15,16} However, training at LML is superior to SML training for producing morphological and neuromuscular adaptations. Therefore, it would be fascinating to compare the effects of submaximal isometric training at LMLs with maximal isometric training at SMLs. As previously mentioned, the body of literature examining the characteristics of "pushing," "holding," and "quasi" isometric actions is growing.^{54-60,78} However, there is a paucity of long-term experimental studies examining these isometric contraction subsets.

5 | PERSPECTIVES

Despite a relatively limited quantity of studies to base conclusions upon, specificity of training applies to isometric

resistance training as it does to traditional dynamic resistance training. Therefore, isometric training should be prescribed in line with the primary outcome goals. Training at LML and with sustained contractions has been found to be beneficial for improving muscle morphology, while high-intensity contractions (>70% MVIC) are likely required to substantially improve tendon structure and function (eg, tendon stiffness). Similarly, ballistic intent has been found to improve rapid force production even though movement velocity is zero. Finally, a greater number of studies, with a broader application of isometric training variations, are needed to determine optimal applications for altering the morphology and improving dynamic performance in athletic, rehabilitative, and special populations alike.

ACKNOWLEDGEMENTS

Dustin J. Oranchuk, Adam G. Storey, André R. Nelson, and John B. Cronin declare that they have no conflicts of interest relevant to the content of this review. No funding was received for this review that may have affected study design, data collection, analysis or interpretation of data, writing of this manuscript, or the decision to submit for publication. Dustin J. Oranchuk was supported by the Auckland University of Technology's Vice Chancellors Doctoral Scholarship.

CONFLICT OF INTEREST

None.

ORCID

Dustin J. Oranchuk  <https://orcid.org/0000-0003-4489-9022>

REFERENCES

- Kraemer WJ, Ratamess NA, French DN. Resistance training for health and performance. *Curr Sports Med Rep*. 2002;1(3):165-171.
- Damas F, Phillips S, Vechin FC, Ugrinowtsch C. A review of resistance training-induced changes in skeletal muscle protein synthesis and their contribution to hypertrophy. *Sports Med*. 2015;45(6):801-807.
- Coupe C, Kongsgaard M, Aagaard P, et al. Habitual loading results in tendon hypertrophy and increased stiffness of the human patellar tendon. *J Appl Physiol*. 2008;105(3):805-810.
- Kongsgaard M, Reitelseder S, Pederson TG, et al. Region specific patellar tendon hypertrophy in humans following resistance training. *Acta Physiol Scand*. 2007;191:111-121.
- Magnusson PS, Kjaer M. The impact of loading, unloading, ageing and injury on the human tendon. *J Physiol*. 2018; <https://doi.org/10.1113/JP275450> [Epub ahead of print].
- Morton SK, Whitehead JR, Brinkert RH, Caine DJ. Resistance training vs. static stretching: effects on flexibility and strength. *J Strength Cond Res*. 2011;25(12):3391-3398.
- Aagaard P, Anderson LJ, Bennekou M, et al. Effects of resistance training on endurance capacity and muscle fiber composition in young top-level cyclists. *Scand J Med Sci Sports*. 2011;21(6):298-307.
- Tanaka H, Swensen T. Impact of resistance training on endurance performance. *Sports Med*. 1998;25(3):191-200.
- Aagaard P, Simonsen EB, Anderson JL, Magnusson P, Dyhre-Poulsen P. Neural adaptation to resistance training: changes in evoked V-wave and H-reflex responses. *J Appl Physiol*. 2002;92(6):2309-2318.
- Kraemer WJ, Ratamess NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc*. 2004;36(4):674-688.
- Burgess KE, Connik MJ, Graham-Smith P, Pearson SJ. Plyometric vs isometric training influences on tendon properties and muscle output. *J Strength Cond Res*. 2007;21(3):986-989.
- Malliaras P, Kamal B, Nowell A, et al. Patellar tendon adaptation in relation to load-intensity and contraction type. *J Biomech*. 2013;46(11):1893-1899.
- Kubo K, Kanehisa H, Fukunaga T. Effects of different duration isometric contractions on tendon elasticity in human quadriceps muscles. *J Physiol*. 2001;536(2):649-655.
- Dietz C, Peterson B. *Triphasic Training: A Systematic Approach to Elite Speed and Explosive Strength Performance*. Hudson, WI: Dietz Sport Enterprise; 2012.
- Hasler EM, Denoth J, Stacoff A, Herzog W. Influence of hip and knee joint angles on excitation of knee extensor muscles. *Electromyogr Clin Neurophysiol*. 1994;34(6):355-361.
- Krebs DE, Staples WH, Cuttita D, Zickel RE. Knee joint angle: its relationship to quadriceps femoris activity in normal and postarthrotomy limbs. *Arch Phys Med Rehabil*. 1983;64(10):441-447.
- Abbott BC, Wilkie DR. The relation between velocity of shortening and the tension-length curve of skeletal muscle. *J Physiol*. 1953;120(1):214-223.
- Tsoukos A, Bogdanis GC, Terzis G, Veligekas P. Acute improvement of vertical jump performance after isometric squats depends on knee angle and vertical jumping ability. *J Strength Cond Res*. 2016;30(8):2250-2257.
- van Beijsterveldt A, van de Port I, Vereijken AJ, Backx F. Risk factors for hamstring injuries in male soccer players: a systematic review of prospective studies. *Scand J Med Sci Sports*. 2013;23(3):253-262.
- Rio E, Kidgell D, Purdam C, et al. Isometric exercise induces analgesia and reduces inhibition in patellar tendinopathy. *Br J Sports Med*. 2015;49(19):1277-1283.
- Rio E, van Ark M, Docking S, et al. Isometric contractions are more analgesic than isotonic contractions for patellar tendon pain: an in-season randomized clinical trial. *Clin J Sport Med*. 2017;27(3):253-259.
- Goodwill AM, Pearce AJ, J KD. Corticomotor plasticity following unilateral strength training. *Muscle Nerve*. 2012;46(3):384-393.
- Murphy AJ, Wilson G, Pryor JF, Newton RU. Isometric assessment of muscular function: the effect of joint angle. *J Appl Biomech*. 1995;11(2):205-215.

24. Murphy AJ, Wilson GJ. Poor correlations between isometric tests and dynamic performance: relationship to muscle activation. *Eur J Appl Physiol.* 1996;77:353-357.
25. Wilson GJ, Murphy AJ. The use of isometric tests of muscular function in athletic assessment. *Sports Med.* 1996;22(1):19-37.
26. Khamoui AV, Brown LE, Nguyen D, et al. Relationship between force-time and velocity-time characteristics of dynamic and isometric muscle actions. *J Strength Cond Res.* 2011;25(1):198-204.
27. McGuigan MR, Newton MJ, Winchester JB, Nelson AG. Relationship between isometric and dynamic strength in recreationally trained men. *J Strength Cond Res.* 2010;24(9):2570-2573.
28. Drake D, Kennedy R, Wallace E. The validity and responsiveness of isometric lower body multi-joint tests of muscular strength: a systematic review. *Sports Med Open.* 2017;3(1):32.
29. Oranchuk DJ, Robinson TL, Switaj ZJ, Drinkwater EJ. Comparison of the hang high-pull and loaded jump squat for the development of vertical jump and isometric force-time characteristics. *J Strength Cond Res.* 2019;33(1):17-24.
30. Alegre LM, Ferri-Morales A, Rodriguez-Casares R, Aguado X. Effects of isometric training on the knee extensor moment-angle relationship and vastus lateralis muscle architecture. *Eur J Appl Physiol.* 2014;114(11):2437-2446.
31. Kubo K, Ohgo K, Takeishi R, et al. Effects of isometric training at different knee angles on the muscle-tendon complex in vivo. *Scand J Med Sci Sports.* 2006;16(3):159-167.
32. Noorkoiv M, Nosaka K, Blazevich AJ. Neuromuscular adaptations associated with knee joint angle-specific force change. *Med Sci Sports Exerc.* 2014;46(8):1525-1537.
33. Noorkoiv M, Nosaka K, Blazevich AJ. Effects of isometric quadriceps strength training at different muscle lengths on dynamic torque production. *J Sports Sci.* 2015;33(18):1952-1961.
34. Schott J, McCully K, Rutherford OM. The role of metabolites in strength training: short versus long isometric contractions. *Eur J Appl Physiol Occup Physiol.* 1995;71(4):337-341.
35. Sterling DR. *Isometric Strength Position Specificity Resulting from Isometric and Isotonic Training as a Determinant in Performance.* Eugene, Ore: University of Oregon; 1969.
36. Rasch PJ, Pierson WR. One position versus multiple positions in isometric exercise. *Am J Phys Med.* 1964;43(1):10-12.
37. Thepaut-Mathieu C, Van Hoeck J, Maton B. Myoelectrical and mechanical changes linked to length specificity during isometric training. *J Appl Physiol.* 1988;64(4):1500-1505.
38. Bandy WD, Hanten WP. Changes in torque and electromyographic activity of the quadriceps femoris muscles following isometric training. *Phys Ther.* 1993;73(7):455-465.
39. Meyers CR. Effects of two isometric routines on strength, size, and endurance in exercised and nonexercised arms. *Res Q Exerc Sport.* 1967;38(3):430-440.
40. Lindh M. Increase of muscle strength from isometric quadriceps exercises at different knee angles. *Scand J Rehabil Med.* 1979;11(1):33-36.
41. Arampatzis A, Karamanidis K, Albracht K. Adaptational responses of the human Achilles tendon by modulation of the applied cyclic strain magnitude. *J Exp Biol.* 2007;210:2743-2753.
42. Arampatzis A, Peper A, Bierbaum S, Albracht K. Plasticity of human Achilles tendon mechanical and morphological properties in response to cyclic strain. *J Biomech.* 2010;43(16):3073-3079.
43. Balshaw TG, Massey GJ, Maden-Wilkinson TM, Tillin NA, Folland JP. Training-specific functional, neural, and hypertrophic adaptations to explosive- vs. sustained-contraction strength training. *J Appl Physiol.* 2016;120(11):1364-1373.
44. Kanehisa H, Nagareda H, Kawakami Y, et al. Effect of equivalent volume isometric training programs comprising medium or high resistance on muscle size and strength. *Eur J Appl Physiol.* 2002;87(2):112-119.
45. Khouw W, Herbert R. Optimisation of isometric strength training intensity. *Aust J Phys Ther.* 1998;44(1):43-46.
46. Szeto G, Strauss GR, De Domenico G, Sun LH. The effect of training intensity on voluntary isometric strength improvement. *Aust J Phys Ther.* 1989;34(4):210-217.
47. Tillin NA, Folland JP. Maximal and explosive strength training elicit distinct neuromuscular adaptations, specific to the training stimulus. *Eur J Appl Physiol.* 2014;114(2):365-374.
48. Maffiuletti NA, Martin A. Progressive versus rapid rate of contraction during 7 wk of isometric resistance training. *Med Sci Sports Exerc.* 2001;33(7):1220-1227.
49. de Ruiter CJ, de Boer MD, Spanjaard M, de Haan A. Knee angle-dependent oxygen consumption during isometric contractions of the knee extensors determined with near-infrared spectroscopy. *J Appl Physiol.* 2005;99:579-586.
50. Sjogaard G, Savard G, Carsten J. Muscle blood flow during isometric activity and its relation to muscle fatigue. *Eur J Appl Physiol Occup Physiol.* 1988;57:327-335.
51. Fisher J, Van-Dongen M, Sutherland R. Combined isometric and vibration training does not enhance strength beyond that of isometric training alone. *J Sports Med Phys Fitness.* 2015;55(9):899-904.
52. Silva HR, Couto BP, Szmuchrowski LA. Effects of mechanical vibration applied in the opposite direction of muscle shortening on maximal isometric strength. *J Strength Cond Res.* 2008;22(4):1031-1036.
53. Alberti G, Ragazzi R. Maximum strength and vertical jump effects of electromyostimulation versus isometric training. *Med Sport (Roma).* 2007;60(4):557-565.
54. Garner JC, Blackburn T, Wiemar W, Campbell B. Comparison of electromyographic activity during eccentrically versus concentrically loaded isometric contractions. *J Electromyogr Kinesiol.* 2008;18(3):466-471.
55. Hunter SK, Ryan DL, Ortega JD, Enoka RM. Task differences with the same load torque alter the endurance time of submaximal fatiguing contractions in humans. *J Neurophysiol.* 2002;88(6):3087-3096.
56. Rudroff T, Barry BK, Stone AL, Barry CJ, Enoka RM. Accessory muscle activity contributes to the variation in time to task failure for different arm postures and loads. *J Appl Physiol.* 2007;102(3):1000-1006.
57. Rudroff T, Justice JN, Holmes MR, Matthews SD, Enoka RM. Muscle activity and time to task failure differ with load compliance and target force for elbow flexor muscles. *J Appl Physiol.* 2011;110(1):125-136.
58. Rudroff T, Kalliokoski KK, Block DE, Gould JR, Klingensmith WC, Enoka RM. PET/CT imaging of age- and task-associated differences in muscle activity during fatiguing contractions. *J Appl Physiol.* 2013;114(9):1211-1219.
59. Schaefer LV, Bittmann FN. Are there two forms of isometric muscle action? Results of the experimental study support a

- distinction between a holding and a pushing isometric muscle function. *BMC Sports Sci Med Rehabil.* 2017;9(11):1-13.
60. Semmler JG, Kornatz KW, Dinunno DV, Zhou S, Enoka RM. Motor unit synchronization is enhanced during slow lengthening contractions of a hand muscle. *J Physiol.* 2002;545(2):681-695.
 61. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med.* 2009;6(7):e1000100.
 62. Brughelli M, Cronin J, Levin G, Chaouachi A. Understanding change of direction ability in sport: a review of resistance training studies. *Sports Med.* 2008;38(12):1045-1063.
 63. Rhea MR. Determining the magnitude of treatment effects in strength training research through the use of the effect size. *J Strength Cond Res.* 2004;18(4):918-920.
 64. Bogdanis GC, Tsoukos A, Methenitis SK, Selima E, Veligeas P, Terzis G. Effects of low volume isometric leg press complex training at two knee angles on force-angle relationship and rate of force development. *Eur J Sport Sci.* 2018;1-9. <https://doi.org/10.1080/17461391.2018.1510989>. [Epub ahead of print].
 65. Young K, McDonagh M, Davies C. The effects of two forms of isometric training on the mechanical properties of the triceps surae in man. *Pflügers Archiv.* 1985;405(4):384-388.
 66. Williams DM. *The Study of Voluntary Activation and Force Production Relationships and Responses to Varied Isometric Strength Training Parameters During Fatiguing and Non-fatiguing Test Protocols.* Iowa City, IA: Iowa Research Online; 2011.
 67. Massey G, Balshaw T, Maden-Wilkinson T, Tillin N, Folland J. **Tendinous tissue adaptation to explosive- vs. sustained-contraction strength training.** *Front Physiol.* 2018;9(1170):1-17.
 68. Waugh CM, Alktebi T, De Sa A, Scott A. Impact of rest duration on Achilles tendon structure and function following isometric training. *Scand J Med Sci Sports.* 2018;28(2):436-445.
 69. Ullrich B, Holzinger S, Soleimani M, Pelzer T, Stening J, Pfeiffer M. Neuromuscular responses to 14 weeks of traditional and daily undulating resistance training. *Int J Sports Med.* 2015;36(7):554-562.
 70. Guex K, Degache F, Morisod C, Saily M, Millet GP. Hamstring architectural and functional adaptations following long vs. short muscle length eccentric training. *Front Physiol.* 2016;7(340):1-9.
 71. Barak Y, Ayalon M, Dvir Z. Transferability of strength gains from limited to full range of motion. *Med Sci Sports Exerc.* 2004;36(8):1413-1420.
 72. Massey CD, Vincent J, Maneval M, Moore M, Johnson JT. An analysis of full range of motion vs. partial range of motion training in the development of strength in untrained men. *J Strength Cond Res.* 2004;18(3):518-521.
 73. Allen TJ, Jones T, Tsay A, Morgan DL, Proske U. Muscle damage produced by isometric contractions in human elbow flexors. *J Appl Physiol.* 2018;124(2):388-399.
 74. Loenneke JP, Pujol TJ. The use of occlusion training to produce muscle hypertrophy. *Strength Cond J.* 2009;31(3):77-84.
 75. Dankel SJ, Mattocks KT, Jessee MB, Buckner SL, Mouser JG, Loenneke JP. Do metabolites that are produced during resistance exercise enhance muscle hypertrophy? *Eur J Appl Physiol.* 2017;117(11):2125-2135.
 76. Schoenfeld BJ, Grgic J, Ogborn D, Krieger JW. Strength and hypertrophy adaptations between low- versus high-load resistance training: a systematic review and meta-analysis. *J Strength Cond Res.* 2017;31(12):3508-3523.
 77. Lasevicius T, Ugrinowitsch C, Schoenfeld BJ, et al. Effects of different intensities of resistance training with equated volume load on muscle strength and hypertrophy. *Eur J Sport Sci.* 2018;18(6):772-780.
 78. Spurway NC. Hiking physiology and the "quasi-isometric" concept. *J Sports Sci.* 2007;25(10):1081-1093.
 79. Akima H, Ryosuke A. Oxygenation and neuromuscular activation of the quadriceps femoris including the vastus intermedius during a fatiguing contraction. *Clin Physiol Funct Imaging.* 2017;37:750-758.
 80. Franchi MV, Reeves ND, Narici MV. Skeletal muscle remodeling in response to eccentric vs. concentric loading: morphological, molecular, and metabolic adaptations. *Front Physiol.* 2017;8(447):1-16.
 81. Magnusson PS, Kjaer M. Region-specific differences in Achilles tendon cross-sectional area in runners and non-runners. *Eur J Appl Physiol.* 2003;90:549-553.
 82. Reeves ND, Narici MV, Maganaris CN. Strength training alters the viscoelastic properties of tendons in elderly humans. *Muscle Nerve.* 2003;28(1):74-81.
 83. Kubo K, Morimoto M, Komuro T, et al. Effects of plyometric and weight training on muscle-tendon complex and jump performance. *Med Sci Sports Exerc.* 2007;39(10):1801-1810.
 84. Arya S, Kulig K. Tendinopathy alters mechanical and material properties of the Achilles tendon. *J Appl Physiol.* 2010;108(3):670-675.
 85. Kubo K, Ishigaki T, Ikebukuro T. Effects of plyometric and isometric training on muscle and tendon stiffness in vivo. *Physiol Rep.* 2017;5(e13374):1-13.
 86. Heinemeier KM, Olesen JL, Haddad F, et al. Expression of collagen and related growth factors in rat tendon and skeletal muscle in response to specific contraction types. *J Physiol.* 2007;582:1303-1316.
 87. Heinemeier KM, Olesen JL, Schjerling P, et al. Short-term strength training and the expression of myostatin and IGF-1 isoforms in rat muscle and tendon: differential effects of specific contraction types. *J Appl Physiol.* 2007;102:573-581.
 88. van Schie HT, de Vos RJ, de Jonge S, et al. Ultrasonographic tissue characterisation of human Achilles tendons: quantification of tendon structure through a novel non-invasive approach. *Br J Sports Med.* 2010;44(16):1153-1159.
 89. Docking SI, Rosengarten SD, Cook J. Achilles tendon structure improves on UTC imaging over a 5-month pre-season in elite Australian football players. *Scand J Med Sci Sports.* 2016;26(5):557-563.
 90. Kubo K, Yata H, Kanehisa H, Fukunaga T. Effects of isometric squat training on the tendon stiffness and jump performance. *Eur J Appl Physiol.* 2006;96(3):305-314.
 91. West DJ, Owen NJ, Jones MR, et al. Relationships between force-time characteristics of the isometric midthigh pull and dynamic performance in professional rugby league players. *J Strength Cond Res.* 2011;25(11):3070-3075.
 92. Timmins RG, Shield AJ, Williams MD, Opar DA. Is there evidence to support the use of the angle of peak torque as a marker of hamstrings injury and re-injury risk. *Sports Med.* 2015;46(1):7-13.
 93. Bowers EJ, Morgan DL, Proske U. Damage to the human quadriceps muscle from eccentric exercise and the training effect. *J Sports Sci.* 2004;22:1005-1014.

94. Brockett CL, Morgan DL, Proske U. Human hamstring muscles adapt to eccentric exercise by changing optimum length. *Med Sci Sports Exerc.* 2001;33(5):783-790.
95. Guex K, Degache F, Gremion G, Millet GP. Effect of hip flexion angle on hamstring optimum length after a single set of concentric contractions. *J Sports Sci.* 2013;31(14):1545-1552.
96. Philippou A, Borgdanis GC, Nevill AM, Maridaki M. Changes in the angle-force curve of human elbow flexors following eccentric and isometric exercise. *Eur J Appl Physiol.* 2004;93(1-2):237-244.
97. Philippou A, Maridaki M, Bogdanis GC, Halapas A, Koutsilieris M. Changes in the mechanical properties of human quadriceps muscle after eccentric exercise. *In Vivo.* 2009;23(5):859-865.
98. Yeung SS, Yeung EW. Shift of peak torque angle after eccentric exercise. *Int J Sports Med.* 2008;29(3):251-256.
99. Philippou A, Maridaki M, Bogdanis GC. Angle-specific impairment of elbow flexors strength after isometric exercise at long muscle length. *J Sports Sci.* 2003;21(10):859-865.
100. Beretić I, Burovic M, Okicik T, Dopsaj M. Relations between lower body isometric muscle force characteristics and start performance in elite male sprint swimmers. *J Sports Sci Med.* 2013;12(4):639-645.
101. Leary BK, Statler J, Hopkins B, et al. The relationship between isometric force-time curve characteristics and club head speed in recreational golfers. *J Strength Cond Res.* 2012;26(10):2685-2697.
102. Haff GG, Carlock JM, Hartman MJ, et al. Force-time curve characteristics of dynamic and isometric muscle actions of elite women olympic weightlifters. *J Strength Cond Res.* 2005;19(4):741-748.
103. Loturco I, Suchomel TJ, Bishop C, Kobal R, Pereira LA, McGuigan M. 1RM measures or maximal bar-power output: which is more related to sport performance? *Int J Sports Physiol Perform.* 2018;1-18. <https://doi.org/10.1123/ijspp.2018-0255>. [Epub ahead of print].
104. Winchester JB, McBride JM, Maher MA, et al. Eight weeks of ballistic exercise improves power independently of changes in strength and muscle fiber type expression. *J Strength Cond Res.* 2008;22(6):1728-1734.
105. Jenkins N, Maramonti AA, Hill EC, et al. Greater neural adaptations following high- vs. low-load resistance training. *Front Physiol.* 2017;8(331):1-15.
106. Balshaw TG, Massey GJ, Maden-Wilkinson TM, et al. Changes in agonist neural drive, hypertrophy and pre-training strength all contribute to the individual strength gains after resistance training. *Eur J Appl Physiol.* 2017;117(4):631-640.
107. Behm DG, Sale DG. Intended rather than actual movement velocity determines velocity-specific training response. *J Appl Physiol.* 1993;74(1):359-368.
108. Kubo J, Chishaki T, Nakamura N, et al. Differences in fat-free mass and muscle thickness at various sites according to performance level among judo athletes. *J Strength Cond Res.* 2006;20(3):654-657.
109. Perez-Gomez J, Redrigeuz GV, Ara I, et al. Role of muscle mass on sprint performance: gender differences? *Eur J Appl Physiol.* 2008;102(6):685-694.
110. Faulkner JA, Davis C, C L, Brooks SV. The aging of elite male athletes: age-related changes in performance and skeletal muscle structure and function. *Clin J Sport Med.* 2008;18(6):501-507.
111. James LP, Haff GG, Kelly VG, Connick MJ, Hoffman BW, Beckman EM. The impact of strength level on adaptations to combined weightlifting, plyometric, and ballistic training. *Scand J Med Sci Sports.* 2018;28(5):1494-1505.
112. Giorgi A, Wilson GJ, Weatherby RP, Murphy AJ. Functional isometric weight training: its effects on the development of muscular function and the endocrine system over an 8-week training period. *J Strength Cond Res.* 1998;21(1):18-25.
113. O'Shea KL, O'Shea JP. Functional isometric weight training: its effects on dynamic and static strength. *J Appl Sport Sci Res.* 1989;3(2):30-33.
114. O'Shea P, O'Shea K, Wynn B. Functional isometric lifting- Part I: theory. *Natl Str Cond Assoc J.* 1987;9(6):44-51.
115. O'Shea P, O'Shea K, Wynn B. Functional isometric lifting-Part II: application. *Natl Str Cond Assoc J.* 1988;10(1):60-62.
116. Muraoka T, Muramatsu T, Fukunaga T, Kanehisa H. Influence of tendon slack on electromechanical delay in the human medial gastrocnemius in vivo. *J Appl Physiol.* 2004;96(2):540-544.
117. Schoenfeld BJ, Peterson MD, Ogborn D, Contreras B, Sonmez GT. Effects of low- vs. high-load resistance training on muscle strength and hypertrophy in well-trained men. *J Strength Cond Res.* 2015;29(10):2954-2963.
118. Moore DR, Young M, Phillips SM. Similar increases in muscle size and strength in young men after training with maximal shortening or lengthening contractions when matched for total work. *Eur J Appl Physiol.* 2012;112(4):1587-1592.
119. Fukunaga T, Kawakami Y, Kubo K, Kanehisa H. Muscle and tendon interaction during human movements. *Exerc Sport Sci Rev.* 2002;30(3):106-110.
120. Kubo K, Morimoto M, Komuro T, Tsunoda N, Kanehisa H, Fukunaga T. Influences of tendon stiffness, joint stiffness, and electromyographic activity on jump performances using single joint. *Eur J Appl Physiol.* 2007;99(3):235-243.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Oranchuk DJ, Storey AG, Nelson AR, Cronin JB. Isometric training and long-term adaptations: Effects of muscle length, intensity, and intent: A systematic review. *Scand J Med Sci Sports.* 2019;00:1–20. <https://doi.org/10.1111/sms.13375>